Module 5:AI Project Implementation

From an approved proposal to a tested, working AI prototype.

Rule #1

Module 5 continues your Module 4 proposal. Implement the project you proposed rather than starting an unrelated one. If the data or scope must change, record what changed and why.

The implementation journey

1. Module 4 Proposal
2. Problem
3. Solution
4. Data
5. EDA
6. Baseline
7. Improve
8. Evaluate
9. Test

What you will produce

Output	Description
Problem definition	Clear explanation of the problem being solved
AI solution design	Chosen task, features, target, workflow, architecture
Dataset and preparation	Documented data source; cleaned, prepared data
EDA	Analysis and visualizations with written interpretation
Baseline and improved models	First working model and justified alternatives
Evaluation and testing	Metrics explained; evidence of whole-system testing
Working prototype	Demonstration of the implemented solution
Documentation	Explanation of the complete implementation

Before you begin: team roles

Review your Module 4 proposal and make sure everyone can answer: What problem are we solving? Who uses the solution? What data do we have? What should the system output? What are the expected limitations?

Stage 1: Understand and Refine the Problem

Do not start with "Which algorithm should we use?" Start with "What problem are we solving, for whom, and what decision or task should the system support?"

Complete this	Guiding question
Problem statement	What is the problem, who is affected, why does it matter, what limits the current approach, and how could AI help?
Target users	Who are the primary and secondary users, and how will they use the system?
Expected input	What information goes in? List each input with its type and description.
Expected output	What does the system produce (class, probability, number, recommendation) and what does it mean to the user?
Success criteria	Measurable targets for model performance, valid-input

	handling, response time and test cases. "The model works" is not a criterion.
Is AI appropriate?	Is there enough relevant data and a learnable pattern? Would a simple rule-based solution do the job?

Input table

Input	Type	Description

Success criteria table

Criterion	Target / expected result
Model performance	
Prediction / output	
User input	
System functionality	
Testing	

Stage 2: Define the AI Solution

Choose an approach that fits the problem, suits your data and team, and is practical for this course.

Task type	Description and example
Classification	Predict a category. Example: will a patient be readmitted, yes or no?
Regression	Predict a number. Example: expected crop yield.
Clustering	Group similar records. Example: customer segments.
Recommendation	Suggest an item or action based on available information.
Generative AI	Generate text or other content. Use only when your approved project needs it, and explain exactly how.

Document the design

- The task you chose and **why**.
- A table of **features** (model inputs) and the **target** (what is predicted).
- The system workflow and a simple architecture.

Variable	Feature or target?	Data type	Purpose

Architecture: User → Interface → Input validation → Preprocessing → Trained model → Prediction → Result
Evidence: AI solution design with feature/target table and architecture diagram.

Stage 3: Collect and Prepare Data

Poor data produces poor models. Understand your data before training anything, and interpret what you see rather than just running commands.

3.1 Document the source

Item	Information
Dataset name	
Source and access method	
Number of records / variables	
Target variable	
Data format	
License / usage information	

3.2 Inspect and check quality

```
df.head(); df.shape; df.info(); df.describe()
df.isnull().sum(); df.duplicated().sum()
df["target"].value_counts(normalize=True)
```

Check for missing values, duplicates, inconsistent categories (Male, male, M), impossible or extreme values, an imbalanced target, and columns that should not be used. Log every problem:

Problem	Evidence	Action taken and reason

3.3 Prepare the data

Step	Guidance
Missing values	Do not automatically drop rows. Choose removal or imputation (median, mode, a category) and justify it.
Duplicates	Record counts before and after removal, and the reason.
Inconsistent data	Standardize categories and record what you changed.
Encoding	Convert categories to numbers with a suitable method (one-hot for unordered, ordinal for ordered).
Scaling	Useful for KNN, SVM, logistic regression and neural networks. Not needed for tree-based models. Record whether you used it and why.

3.4 Split the data and avoid leakage

A typical split is 80% training and 20% testing with a fixed random state; use stratified splitting for imbalanced classes. Record your split and the reason.

3.5 Preprocessing record

Step	Action	Reason
Missing values		

Duplicates		
Encoding		
Scaling		
Outliers		
Data split		

Evidence: prepared dataset and preprocessing documentation.

Stage 4: Exploratory Data Analysis

EDA answers useful questions. Every chart needs a purpose and a written interpretation, never decoration.

- **Variables:** what each represents, its type, its range, and whether it has missing values.
- **Descriptive statistics:** mean, median, minimum, maximum, standard deviation. Explain what they tell you.
- **Distributions:** histograms, box plots, bar charts. Are they balanced? Are there extreme values?
- **Relationships:** feature versus target, correlation matrix, group comparisons.

Finding	**Evidence**	**Possible impact on modeling**

Close with an EDA summary covering what you learned, which variables look important, which quality issues you found and fixed, and what to keep in mind while modeling (imbalance, leakage risk, correlated features).

Evidence: EDA notebook section with visualizations and written findings.

Stage 5: Build a Baseline Model

A baseline is your first simple working model, the benchmark later models must beat. Start simple.

Task	Beginner-friendly algorithms
Classification	Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, Naive Bayes
Regression	Linear Regression, Decision Tree Regressor, Random Forest Regressor
Clustering	K-Means

- **Justify your choice:** why it fits the problem and data, why it is practical, and what its limits are. "It is popular" is not a reason.
- Compare against a **trivial reference** (for example always predicting the majority class). If your model barely beats it, say so.
- Keep features, target, training data and test data clearly separated. Train, predict on the test set, and record the first results.

Model	Metric 1	Metric 2	Metric 3	Notes
Baseline				

Evidence: working baseline model with initial evaluation.

Stage 6: Train and Improve the Model

Better does not only mean a higher score. A good model is also reliable, understandable, efficient and suitable for its use.

- **Compare two or three sensible models** using the same data and evaluation method. Do not try algorithms without a reason.
- **Basic tuning only:** tree depth, number of trees, number of neighbors, regularization. Use cross-validation and log each change.
- **Handle class imbalance if present:** class weights or resampling, applied to training data only.
- **Check overfitting:** compare training and validation performance. 98% versus 71% needs investigation.
- **Record failed experiments:** what you tried, what happened, what you learned.

Model comparison

Model	Accuracy / MAE	Precision / RMSE	Recall / R^2	F1	Notes
Model 1					
Model 2					
Model 3					

Tuning log

Parameter	Before	After	Reason

Choose the final model by weighing performance, consistency, complexity, interpretability, speed and suitability, not just the highest number. Record the selected model and why.

Evidence: model comparison table and justified final-model decision.

Stage 7: Evaluate the Model

How well does the model perform on data it has never seen? Evaluate the final model once, on the untouched test set, and explain every number.

Metric	What it tells you
Accuracy	Share of predictions that are correct. Best when classes are balanced and error costs are similar.
Precision	Of cases predicted positive, how many were actually positive.
Recall	Of actual positive cases, how many the model found.
F1-score	Combined measure of precision and recall.
Confusion matrix	True/false positives and negatives; shows which errors occur.
ROC-AUC	How well the classifier separates classes across thresholds.
MAE (regression)	Average absolute difference between predicted and actual values.
RMSE (regression)	Like MAE but penalizes larger errors more strongly.
R^2 (regression)	How much variation in the target the model explains.

- **Interpret in context:** "An F1-score of 0.82 shows a reasonable balance on this test sample, within the limits of this dataset." Never "our model is 82% perfect."
- **Read the confusion matrix:** which cases are misclassified, which error is most common, and why might that be?
- **Look at errors:** examine a few wrong predictions and any group where the model does noticeably worse.

Metric	Result	What it means for this project

Evidence: final results, interpretation and limitations.

Stage 8: Test the Complete Solution

A model that works in a notebook can still fail in a real system. Test the whole chain: input, validation, preprocessing, model, prediction, output.

Test case	Input	Expected	Actual	Pass / Fail
Normal, complete input				
Missing value		Clear message		
Invalid number or format		Rejected safely		
Unexpected category		Handled		
Extreme value		Handled or warned		

Final Documentation and Submission

Report structure

1. Project title
2. Problem statement
3. Target users
4. Proposed AI solution
5. Dataset
6. Data preparation
7. Exploratory data analysis
8. Baseline model
9. Model improvement
10. Final model
11. Evaluation
12. Testing
13. Working prototype
14. Limitations and responsible-use notes
15. Future improvements
16. Conclusion
17. Team contributions

Group reflection

What did we learn? What was the most difficult part? What technical problem did we encounter and how did we solve it? What would we improve with more time? What did we learn about working as a team?

The goal of Module 5

It is not to create a perfect AI system. It is to show that your group can take an approved idea and implement it systematically: understand the problem, define the solution, prepare the data, explore, build a baseline, improve, evaluate and test. Every member should understand the work submitted.